

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (Bsc.IT Semester III)
Data Structures Practical

Practical - V

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Class : SYIT	Batch : 1
Date of Assignment : 22-7-26	Date/Time of Submission : 22-7-26

Binary Search Tree: Write a program to:

- a. Create a binary search tree.

Code:-

```
class Treenode:
    def __init__(self, data):
        self.data = data
        self.left = None
        self.right = None
```

- b. Insert nodes into a binary search tree.

Code:-

```
root = Treenode('R')
nodeA = Treenode('A')
nodeB = Treenode('B')
nodeC = Treenode('C')
nodeD = Treenode('D')
nodeE = Treenode('E')
nodeF = Treenode('F')
nodeG = Treenode('G')

root.left = nodeA
root.right = nodeB

nodeA.left = nodeC
nodeA.right = nodeD

nodeB.left = nodeE
nodeB.right = nodeF

nodeF.left = nodeG

print("root.right.right.left.data: ", root.right.right.left.data)
print("Panvir, S031")
```

Output:-

```
===== RESTART: D:/  
root.right.right.left.data:  G  
Panvir, S031
```

c. Search for a node in a binary search tree.

Code:-

```
def search(root, key):  
    if root is None:  
        return False  
    print("checking: ", root.data)  
  
    if root.data==key:  
        return True  
  
    return search(root.left, key) or search(root.right, key)  
  
key=input("enter element to search: ")  
  
#search  
if search(root, key):  
    print("element found")  
else:  
    print("Element not found")  
print("Panvir, S031")
```

Output:-

```
enter element to search: G  
checking:  R  
checking:  A  
checking:  C  
checking:  D  
checking:  B  
checking:  E  
checking:  F  
checking:  G  
element found  
Panvir, S031
```

Curiosity Questions:

1. In a BST, which side (left or right) has **smaller numbers** than the root?
2. If we insert numbers 10, 20, 5 into a BST, which number will be at the root?
3. What will the inorder traversal of a BST give — sorted numbers or random numbers?
4. Where will the **largest number** in a BST always be found?
5. If the root of a BST is 50 and we search for 70, will we go left or right? Why?
6. If a BST has only **one node**, what is its inorder traversal?
7. Can we create a BST from characters like A, B, C instead of numbers?
8. If we insert numbers 5, 10, 15, 20 in order, will the tree be balanced or like a straight line?
9. In a BST, do we check the left child first or the right child first in inorder traversal?
10. If we delete the root of a BST, will the tree disappear?